REVIEW 1

Design of a Temperature-Compensated PLL Frequency Synthesizer

An Open-Source sky130 Design and Verification Flow, Cross-Validated Against Commercial EDA

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Project Guide
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1 Executive Summary

This report documents Review 1 of a project originally scoped as a Cadence Virtuoso/Spectre schematic-level PLL design (see the Zeroth Review deck, *Frequency Synthesizer Presentation.pdf*). Two changes were made before implementation began, both driven by practical constraints and both strengthening the project rather than diluting it:

- **Tooling**: without a local Cadence license, the design flow was rebuilt entirely on open-source tools — Xschem-style hand-written SPICE netlists, ngspice for simulation, Magic and KLayout reserved for the physical-design stage, all running headless (no GUI) on the SkyWater sky130 open PDK.
- **Scope**: the original brief was schematic-and-simulation only, with no layout. This plan adds a full schematic-to-GDS flow, and reframes the project's contribution around a genuine, literature-verified research question rather than a reproduction of known circuit topologies.

Review 1 targeted 50% completion: a working closed-loop PLL, transistor-level for every block except a deliberately idealized charge pump, simulated and characterized against a derived analytical model. **That target was met.** The loop locks in 577 ns against a 2 µs specification, matching an independent averaged behavioural model (573 ns). Two real simulation-setup bugs were found and fixed along the way (Section 7), and two specification inconsistencies inherited from the original brief were caught, diagnosed, and resolved (Section 8).

Erratum (2026-10-11). As first issued, this report stated that the loop filter gave $f_n \approx 1.98$ MHz, $\zeta \approx 0.72$, phase margin 70–72° and a "450 ns" predicted lock time, that $K_{VCO} \approx 670$ MHz/V, and that the VCO "does not sustain oscillation" below $\approx$ 0.6–0.65 V. All three were wrong and are corrected in place below: the loop-filter formula omitted a factor $1/2\pi$ (correct: $f_n \approx 0.9$ MHz, $\zeta \approx 0.33$, PM $\approx$ 30–33°, i.e. underdamped); the local K_{VCO} at 0.9 V is $\approx$ 890 MHz/V; and the VCO oscillates (slowly) well below 0.65 V, so there is no hard floor. The simulated results (577 ns lock, tuning curve, tempcos) are unchanged. See `docs/loop_model.md` and `STATUS.md`.

Finding

The project's central research contribution is not the PLL circuit itself (a temperature-compensated ring-oscillator PLL is known, published silicon) but a quantified study of whether open-source EDA model fidelity survives *cancellation-based* designs — where a correction network subtracts two large temperature-dependent quantities to leave a small residual, amplifying simulator/model disagreement by roughly the compensation ratio. This is detailed in Section 2.

2 Research Framing and Novelty

2.1 The problem with the inherited specification

The Zeroth Review's headline target — "±0.15 ppm frequency stability across the temperature range" — is not physically meaningful for a reference-locked PLL. In lock, $f_{\text{out}} = N \cdot f_{\text{ref}}$ exactly; temperature does not move it while the loop stays locked, so ppm-level output stability is inherited from the reference crystal, not from anything the PLL itself contributes. What temperature actually degrades is: the VCO's free-running frequency (hence the required control voltage and lock range),

K_{VCO} (hence loop bandwidth and phase margin), lock time, and phase noise. The specification has been restated accordingly (Section 8).

2.2 What is *not* novel here, stated plainly

- A PTAT/CTAT-compensated ring-oscillator VCO inside a PLL: fabricated and published in 180 nm CMOS (Souliotis et al., *AEU Int. J. Electron. Commun.*, 2022), achieving $-1.5\%/+0.4\%$ over -40 to 120°C.
- A PLL built with open-source tools in sky130: multiple existing repositories (e.g. the VSD-internship `apll_01v8_sky130` project, several TinyTapeout PLL submissions).
- A fully open-source PLL with an LC-VCO: Thiriloganathan et al. (SMACD 2026) built a 2.4 GHz fractional-N LC-VCO PLL in the IHP SG13G2 PDK using Xschem/ngspice/KLayout/OpenEMS.
- "Open-source EDA compares reasonably to commercial tools": established repeatedly for digital logic, and partially for analog.

2.3 The gap this project occupies

The closest prior work, Freitas et al. (TWEPP 2025, `arXiv:2512.06122`), compares an open-source flow (Xschem+ngspice) against a commercial one (Virtuoso+Spectre) in the IHP SG13G2 PDK across three blocks, one of which is a differential current-starved ring-oscillator VCO — close to this project's own VCO benchmark. Critically, their published results report the open-source flow's numbers for the VCO *alone*; no quantified open-vs-commercial comparison exists for that block, only qualitative statements ("required more manual tuning," "a slightly wider spread in oscillation metrics"). Their VCO is also open-loop and has no temperature compensation, no PVT sweep, and no Monte Carlo characterization.

	Quantified open-vs-commercial	Closed-loop PLL	Temperature compensation	PVT + Monte Carlo
Freitas et al. 2025	No (qualitative)	No	No	Digital blocks only
Thiriloganathan et al. 2026	No	Yes	No	No
This project	**Yes (planned)**	**Yes**	**Yes**	**Yes (planned)**

2.4 The falsifiable hypothesis

Temperature compensation is a *cancellation* problem: a correction network is sized so its temperature coefficient nearly cancels the VCO's own, leaving a residual drift one to two orders of magnitude smaller than either term. In such problems, simulator/model disagreement does not pass through proportionally — it is amplified by roughly the compensation ratio.

> **H1:** Let ε_{raw} be the relative cross-flow (ngspice-vs-Spectre) disagreement on directly simulated quantities, and C the compensation ratio (uncompensated drift / compensated residual drift). Then $\varepsilon_{\text{res}} \approx C \cdot \varepsilon_{\text{raw}}$ on the compensated residual — implying a trust ceiling $C_{\max} \approx 1/\varepsilon_{\text{raw}}$ beyond which an open-source flow cannot be relied on to design a correction.

Both outcomes of this test are publishable: confirmation gives the open-silicon community a concrete, transferable design rule; refutation is stronger evidence for open-source precision-analog viability than anything currently published. Neither outcome depends on the compensation *circuit* being novel, which deliberately protects the thesis from being scooped on circuit design alone.

3 Tooling and Environment

Virtuoso component	**Open-source replacement**
Schematic Editor	Hand-written SPICE (Xschem-format netlists)
Spectre (transient/AC)	ngspice-42
ADE Explorer	ngspice batch mode + Python post-processing
ADE Assembler (corners/MC)	CACE (planned, Review 2)
Spectre RF PSS/Pnoise	No open equivalent — transient/jitter estimate
Layout Suite	Magic 8.3.105 + KLayout 0.28.16 (Review 3)
Assura/Calibre LVS	netgen (tooling gap identified, see Section 9)
PDK	SkyWater sky130B via `volare`, pinned commit

Every tool runs headless from the command line: ngspice in `-b` batch mode, Python (no GUI backend) for waveform post-processing and plotting. The sky130 PDK was pulled via `volare` and pinned to a fixed commit hash for reproducibility. One tooling gap was found: Ubuntu's `netgen` package is the TU Vienna finite-element mesh generator, not Tim Edwards' IC-design LVS tool of the same name — flagged for Review 3, not yet blocking.

4 Block 1: Voltage-Controlled Oscillator

4.1 Topology

A 5-stage current-starved ring oscillator in sky130, with a diode-connected NMOS/PMOS bias generator converting V_{ctrl} into a mirrored bias current for every stage's PMOS current source, and V_{ctrl} driving each stage's NMOS current source directly.

4.2 An adversarial review pass, and what it changed

Before any number from the first simulation pass was trusted, the netlist, testbenches, and results were handed cold to an independent reviewer instance instructed to critique the work as a skeptical thesis-committee member would. It returned five must-fix findings, not style notes:

1. Amplitude extraction (`v_min`/`v_max`) was computed on the pre-skip window, contaminating a "swings beyond the rails" claim with startup transient.
2. Cold-start initial conditions had all 5 ring stages switching synchronously from an identical state — a textbook way to manufacture spurious ringing.
3. A 4-period frequency estimate was reported with the same apparent confidence as a 99-period one.
4. Supply current — an explicit checklist deliverable — was simply missing.

5. K_{VCO} and the temperature coefficient were both single two-point secants off a curve later shown to be measurably nonlinear, which would have silently contaminated the compensation-ratio measurement this whole project depends on.

All five were fixed (staggered initial conditions, post-skip-only amplitude extraction, a low-confidence flag below 10 periods, an added current-sense measurement, and polynomial fits replacing secants) before any VCO number below was accepted.

4.3 Results

Quantity	Value	Caveat
K_{VCO} (local derivative @ 0.9 V, tt, 27°C)	≈890 MHz/V (887)	Strongly nonlinear curve; corrected from 670 (Erratum)
Uncompensated tempco, tt (secant, $-40 \rightarrow 85$°C)	+681 ppm/°C	Local slope varies 3.4× across range
Uncompensated tempco, ss corner	+1070 ppm/°C	
Uncompensated tempco, ff corner	+332 ppm/°C	**> 3× spread across corners**
Low-V_{ctrl} behaviour	no hard floor (18 MHz @ 0.65 V)	Device-threshold-limited (Erratum)

Finding

The uncompensated temperature coefficient varies by more than 3× across process corners (332 ppm/°C at ff versus 1070 ppm/°C at ss). A compensation network sized against the typical corner alone would be wrong by more than 3× at a real corner.

Open Risk

The VCO does not sustain oscillation down to the originally specified V_{ctrl} floor (0.27 V, i.e. $0.15 \cdot V_{DD}$). The ring does not stop oscillating at a sharp threshold (corrected 2026-10-11; the first issue of this report read a short-simulation-window artifact as oscillation being lost): f falls from 18 MHz at 0.65 V to 6.5, 2.1 and 0.6 MHz at 0.60, 0.55 and 0.50 V, against a 226 MHz target, so ≈ 0.65 V is the practical floor. This is for the *best-behaved* corner/temperature — cold temperature and the ss corner can only push the practical boundary higher. This is a genuine contradiction between a locked specification and measured data, resolved in Section 8.

5 Blocks 2–4: PFD, Charge Pump, Loop Filter, Divider

5.1 Phase-Frequency Detector

Built from rising-edge pulse generators feeding a pair of cross-coupled NOR SR-latches, rather than the more commonly diagrammed resettable-D-flip-flop structure. An async-resettable transmission-gate flip-flop was designed first and rejected: reliably forcing its stored state without fighting its own bistable feedback loop proved fiddly, while the edge-detector/SR-latch structure is a standard, simpler alternative with identical external behavior. Verified functionally by sweeping REF and DIV at slightly different frequencies (20 MHz vs. 18 MHz) so their relative phase drifts through

a full cycle within one simulation, confirming UP/DN pulse widths track the phase relationship correctly in both directions.

5.2 Charge Pump — a deliberate simplification

Modeled behaviorally (an ideal current source gated by UP/DN) rather than at transistor level. This is standard PLL design sequencing: verify loop dynamics against the analytical model with an ideal charge pump first, and only invest in transistor-level switches, current-mirror matching, and charge-injection modeling once mismatch is something the project is actually measuring — which is Review 2 (Monte Carlo) territory, so no effort is wasted.

5.3 Loop Filter

A passive lead-lag network (R2 + C1 series, C2 shunt for ripple), with component values derived from a standard second-order charge-pump-PLL analytical model (Section 6).

5.4 Feedback Divider

A ÷8 chain of three toggle flip-flops. Built from a transmission-gate D-latch design that was hand-verified *before* first simulation: an initial latch design used one inverter plus a bare pass gate in the storage loop, which gives only one inversion around the loop (astable, not bistable) — a genuine design error caught by re-deriving the topology from first principles rather than by debugging a broken waveform. The corrected two-inverter design was verified functionally: measured division ratio 8.004 against a target of 8.000.

6 Analytical Loop Model

Using the measured K_{VCO} (670 MHz/V as first reported; corrected local value at 0.9 V is ≈ 890 MHz/V), a nominal VCO operating point of 226.2 MHz at $V_{\text{ctrl}} = 0.9$ V, a divider ratio $N = 8$ (giving $f_{\text{ref}} = 28.275$ MHz), and a charge pump current $I_{cp} = 20\,\mu$A, standard second-order type-II charge-pump-PLL design equations were solved for a target natural frequency $f_n \approx 2$ MHz and damping $\zeta \approx 0.71$:

$$\omega_n^2 = \frac{I_{cp} \cdot K_{VCO}}{2\pi \cdot N \cdot C_1} \qquad\qquad \zeta = \frac{R_2 C_1 \omega_n}{2}$$

(K_{VCO} in rad/s/V; the $1/2\pi$ is the phase-detector/charge-pump gain $I_{cp}/2\pi$ per radian.) The values chosen were $C_1 = 68$ pF, $R_2 = 1.7$ kΩ, and $C_2 = C_1/10 = 6.8$ pF (standard ripple-pole placement).

Erratum (2026-10-11). The design was originally computed without the $1/2\pi$ in ω_n^2, which inflated ω_n by $\sqrt{2\pi}$ and gave $f_n \approx 1.98$ MHz, $\zeta \approx 0.72$, loop bandwidth ≈ 2.85 MHz, phase margin 70–72° and a "450 ns" lock prediction. Recomputed correctly for these rounded values at the measured $K_{VCO} = 887$ MHz/V: $f_n \approx 0.91$ MHz, $\zeta \approx 0.33$, loop bandwidth ≈ 0.6 MHz, phase margin ≈ 30–33° (100ζ rule; C_2 erodes it slightly more). The loop as built is underdamped, which the simulated overshoot (to ≈ 0.92 V from a 0.85 V start) confirms; it still meets the $< 2\,\mu$s specification. The "450 ns" figure is withdrawn: it agreed with the measured lock time only by coincidence. A retune of R_2 (about 3.6 kΩ for $\zeta \approx 0.7$, not simulated) is a Review 3 item.

7 Closed-Loop Lock Demonstration

The full loop (REF → PFD → Charge Pump → Loop Filter → VCO → ÷8 → back to PFD) was assembled and simulated at the tt corner, 27°C, starting from $V_{\text{ctrl}} = 0.85\,\text{V}$ (a "coarse-tuned start," well inside the VCO's usable range — see the risk discussion below).

Quantity	Measured	Target / Prediction
Lock time (settles within ±2% of final value)	**577 ns**	$< 2\,\mu$s (spec); 573 ns (independent averaged model)
Settled V_{ctrl}	0.902 V	0.9 V design point
Settled divider output frequency	28.515 MHz	28.275 MHz (0.85% high)
Settled VCO output frequency	227.96 MHz	226.2 MHz (0.78% high)

The small residual frequency offset (≈0.8%) is consistent with normal charge-pump/PFD limit-cycle ripple in a real type-II CP-PLL. The measured 577 ns agrees to under 1% with an independent averaged (phase-domain) behavioural model of the same loop run on the measured tuning curve (573 ns, `scripts/loop_check.py`). (Erratum, 2026-10-11: the first issue of this report explained a 577 vs. 450 ns gap by PFD reset delay and discrete pumping; the 450 ns figure was wrong and that explanation is withdrawn.)

7.1 Two real bugs found and fixed

SPICE current-source polarity. The first attempt at the charge pump produced a control voltage that ramped monotonically negative regardless of UP/DN. Rather than guess at a fix against the full three-minute closed-loop simulation, the ambiguity was resolved with an isolated four-line test: charging a bare capacitor with a behavioral current source confirmed that ngspice's `B`-element convention (`B n1 n2 I=expr`) draws current *out of* `n1` when positive — the opposite of the naive reading. The node order was corrected accordingly.

Inconsistent capacitor initial condition. After fixing the polarity, the loop still collapsed from its starting control voltage down to ≈0.1 V within the first 40 ns on every run. The first hypothesis — that ∼30 unstaggered digital gate nodes in the PFD/divider were producing a startup glitch — looked reasonable and was tested by adding a 50 ns charge-pump hold gate. *The collapse still happened, in full, inside that 50 ns window*, which is what disproved the hypothesis rather than confirmed it: if the charge pump was provably silent and the collapse still occurred, the charge pump could not be the cause. The actual cause was simpler and easy to miss: an `.ic` was set on the control-voltage node but not on the loop filter's own internal node, which defaulted to 0 V. With the charge pump idle, the two loop-filter capacitors simply redistributed their mismatched initial charge to $V_{\text{final}} = C_2 V_0/(C_1 + C_2) \approx 0.077\,\text{V}$ — matching the observed collapse almost exactly. The fix was a one-line addition of a consistent initial condition.

Open Risk

The accepted lock result assumes the loop starts already inside the VCO's confirmed-oscillating range. A genuine cold start from 0 V drives the charge pump into a multi-microsecond constant-current integration phase before the VCO begins oscillating at all and the divider starts toggling — directly compounding the VCO lock-range finding from Section 4. This is not yet solved; real PLLs typically address it with a separate frequency-acquisition-aid circuit, which is out of scope for Review 1 and flagged for Review 2.

8 Decisions Made This Review

Decision

PDK: SkyWater sky130, not the originally specified Cadence GPDK. GPDK is NDA-bound and cannot exist outside a Virtuoso seat. sky130 is the only open PDK with an officially sanctioned Cadence Education Kit / ADK, which the eventual Virtuoso cross-validation (Review 3) structurally requires.

Decision

$V_{\mathbf{ctrl}}$ lock-range floor revised from $[0.15 \cdot V_{DD}, 0.85 \cdot V_{DD}] = [0.27\text{V}, 1.53\text{V}]$ to $[0.35 \cdot V_{DD}, 0.85 \cdot V_{DD}] \approx [0.63\text{V}, 1.53\text{V}]$. The original floor is unusable for this topology because it is device-threshold-limited (the NMOS current source's gate is driven directly by V_{ctrl}; the VCO is under 18 MHz below ≈ 0.65 V — not a hard oscillation cutoff, see the Erratum in the Executive Summary), not a sizing oversight; reaching lower would need a fundamentally different level-shifted control scheme, not justified by the specs in play.

Decision

Compensation ratio C locked as *per-corner*, $C(\text{corner})$, rather than a single scalar — a direct consequence of the $> 3\times$ tempco spread measured across ff/tt/ss.

Decision

Frequency stability target restated from "±0.15 ppm" (not physically meaningful for a locked PLL) to a residual VCO free-running drift target of $< \pm 0.3\%$ **($\approx \pm 24$ ppm/°C)** over −40 to +85°C, grounded against comparable published silicon (Souliotis et al. 2022: $-1.5\%/+0.4\%$ in 180 nm).

9 Open Risks

- **Virtuoso/Cadence sky130 ADK access** is the single highest-severity risk: the entire cross-validation half of the research question depends on it, and it has not yet been confirmed with the department. This should be resolved before Review 2 concludes, not discovered as a blocker at Review 3.
- **Cold-start acquisition** (Section 7) is unresolved: either design a fast-acquisition charge-pump mode, or formally document the coarse-tuned-start assumption as a stated design constraint.
- **netgen tooling gap**: the correct IC-design LVS tool needs to be built from source or obtained via a container image before Review 3's LVS step.
- **PVT coverage** on the closed loop is currently a single point (tt, 27°C); the full corner×temperature sweep is Review 2 scope via CACE.

10 Roadmap to Review 2 and Review 3

Review 2 (target 75%): temperature-compensation network design targeting the per-corner $C(\text{corner})$ definition; full 5-corner PVT sweep via CACE; Monte Carlo mismatch characterization; a transistor-level charge pump (needed once mismatch is being measured); an ngspice-vs-Xyce cross-check as an independent-codebase parity measurement; a jitter-derived phase-noise estimate; and a figure-of-merit-normalized benchmark against the five references from the original literature survey.

Review 3 (target 100%): full layout in Magic with DRC closure, LVS via netgen, post-layout re-simulation, the Cadence/Virtuoso cross-validation arm (the schematic rebuilt independently against the sky130 Cadence ADK, run through true Spectre RF PSS/Pnoise analysis), the quantified parity tables (ε_{raw}, ε_{res}, C) that test the central H1 hypothesis, and a paper draft targeting a workshop venue such as SMACD or a FOSSi-affiliated track.

11 References

1. Freitas, E. M. V. et al., "Using Open Source EDA Tools in ASICs for HEP: A Mixed Comparison," TWEPP 2025 / `arXiv:2512.06122`.
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5. "A 12GHz 67% Tuning Range, 0.37ps RJrms PLL with LC-VCO Temperature Compensation Scheme in 0.13µm CMOS," IEEE, 2014.
6. "Design of an Integrated CMOS PLL Frequency Synthesizer," IEEE Xplore.
7. "Temperature and Power Supply Compensated CMOS Clock Circuit Based on Ring Oscillator," *Electronics (MDPI)*, vol. 12, no. 3, 2023.
8. "Phase-Locked Loop (PLL)-Based Frequency Synthesizer for Digital Systems Driving," Cadence Virtuoso GPDK 45nm implementation, ResearchGate.

Full source, testbenches, raw simulation data, and the living project charter are maintained in the project repository (`PROJECT_CHARTER.md`, `pll-sky130/STATUS.md`, and per-block `FINDINGS.md` files).